

Lab 5 – Convolution

EE252L/CE251L: Signals and Systems Lab

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Task 1

1) Does the order of inputs matter? Is convolution always commutative, i.e. is $u(t) * v(t) = v(t) * u(t)$?

A) yes the convolution is commutative because if even we interchange the both functions $u(t)$ and $v(t)$ the convolution output stays the same

2) Describe your general observations about the convolution output w compared to the inputs u and v ?

A) I observed by looking at the output of both functions $u(t)$ and $v(t)$ that the output that came was the magnitude of area under both $u(t)$ and $v(t)$ curves at that point in time when one of the function is fixed in time and the other function is at t and the convolution was maximum when t becomes equal to (the coordinates of fixed function in time axis)

Task 2

1) How is the length of convolution output related to the length L of the two input signals?

A) the length of convolution is equal to $(2L)-1$ where L is the length of both functions

2) What do you think will happen if the input sequences had different lengths? say, M and N , instead of L .

A) If we take the length of each function as M and N the length of the convolutions becomes $(M+N) - 1$

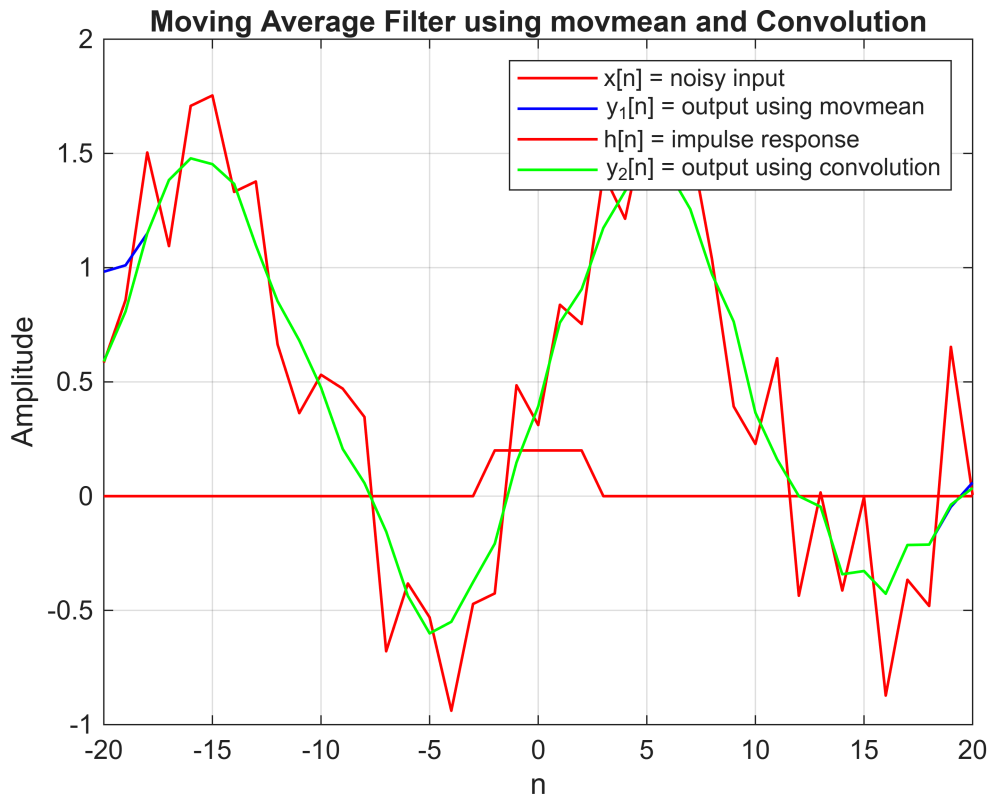
Task 3

```
clc;
clear close all;
n = -20:20;
z = sin(0.3142*n);
noise = rand(size(n));
x = z + noise;
y1 = movmean(x, 5);
delta = (n==0);
h = movmean(delta, 5);
y2 = conv(x, h, "same");
figure;
plot(n, x, 'r-', 'LineWidth', 1);
hold on;
plot(n, y1, 'b-', 'LineWidth', 1);
plot(n, h, 'r-', 'LineWidth', 1);
plot(n, y2, 'g-', 'LineWidth', 1);
```

```

grid on;
xlabel('n');
ylabel('Amplitude');
title('Moving Average Filter using movmean and Convolution');
legend('x[n] = noisy input', 'y_1[n] = output using movmean', 'h[n] = impulse response', 'y_2[n] = output using convolution');

```



Task 4

• prove of linearity: for two inputs $x_1[n]$ & $x_2[n]$

$$T(x_1[n] + x_2[n]) \\ = \frac{1}{5} \sum_{k=0}^4 (x_1[n-k] + x_2[n-k])$$

$$= y_1[n] + y_2[n]$$

so, additivity property is satisfied.

• homogeneity:

$$T(ax[n]) = \frac{1}{5} \sum_{k=0}^4 ax[n-k] = a \frac{1}{5} \sum_{k=0}^4 x[n-k] \\ = ay[n]$$

hence the average filter is linear.

Task 5

```
clc;
clear close all;

Fs = 1000;
t = 0: 1/Fs : 1;
f0 = 5;
x = sin(2*pi*f0*t);
```

```

h1 = ones(1,5)/5;

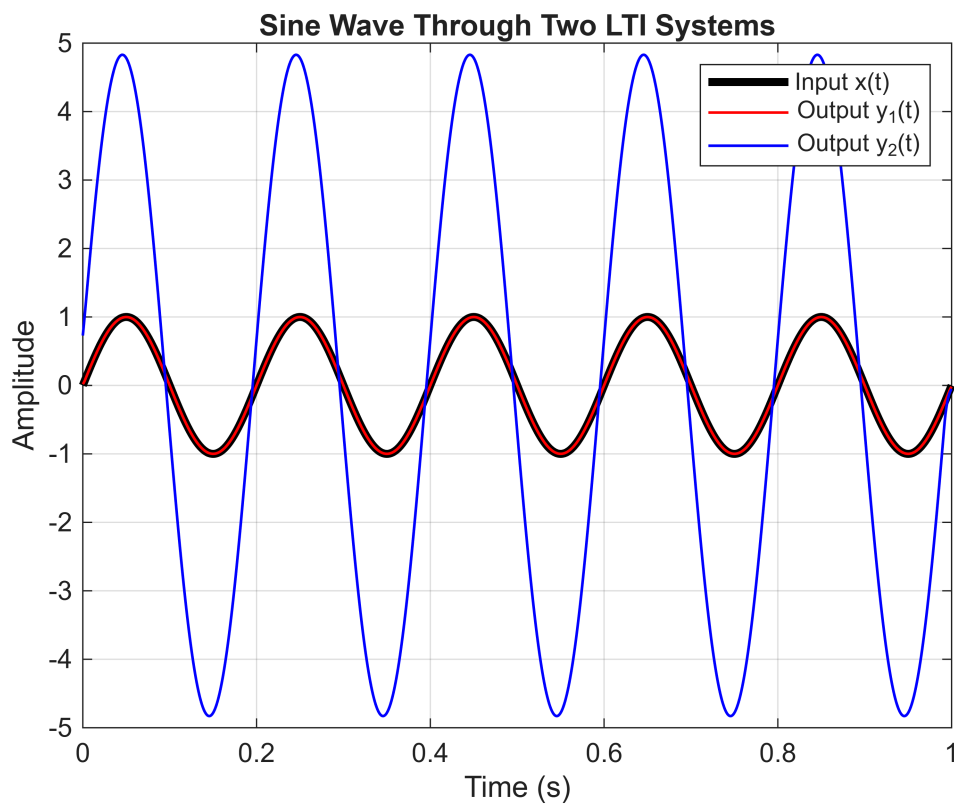
n = 0:15;
h2 = (0.8).^n;

y1 = conv(x, h1, 'same');
y2 = conv(x, h2, 'same');

figure;
plot(t, x, 'k', LineWidth=3);
hold on;
plot(t, y1, 'r', LineWidth=1);
plot(t, y2, 'b', LineWidth=1);
hold off;

grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Sine Wave Through Two LTI Systems');
legend('Input x(t)', 'Output y_1(t)', 'Output y_2(t)');

```



```

[pks1, locs1] = findpeaks(y1, t);
T1 = locs1(2) - locs1(1);
f1 = 1/T1;

[pks2, locs2] = findpeaks(y2, t);
T2 = locs2(2) - locs2(1);

```

```
f2 = 1/T2;
```

```
disp("Input Frequency: " + f0)
```

Input Frequency: 5

```
disp("System 1:");
```

System 1:

```
disp("Time Period T1: " + T1)
```

Time Period T1: 0.2

```
disp("Frequency f1: " + f1)
```

Frequency f1: 5

```
disp("System 2:");
```

System 2:

```
disp("Time Period T2: " + T2)
```

Time Period T2: 0.2

```
disp("Frequency f2: " + f2)
```

Frequency f2: 5

Task 6

```
[h, Fh] = audioread("/MATLAB Drive/lab  
5/1st_baptist_nashville_balcony.wav");  
disp(size(h))
```

288000

2

```
soundsc(h,Fh) % Play the sound  
pause(length(h)/Fh) % Wait for audio to finish playing  
  
[x, Fx] = audioread("/MATLAB Drive/lab 5/ringtone.wav"); % Read input  
disp(size(x))  
soundsc(x,Fx) % Play the sound  
pause(length(x)/Fx) % Wait for audio to finish playing  
  
h1 = h(:,1); % Extract the first channel from the impulse response  
%In simple words extracting all the rows of first column  
h2 = h(:,2); % Extract the second channel from the impulse response  
%ISimilarly extracting all the rows of second column of h  
  
y1 = conv(h1, x);  
y2 = conv(h2, x);
```

```

% Concatenate into stereo output
y = [y1, y2];

% Play the reverberated sound
soundsc(y, Fx)
pause(length(y)/Fx)

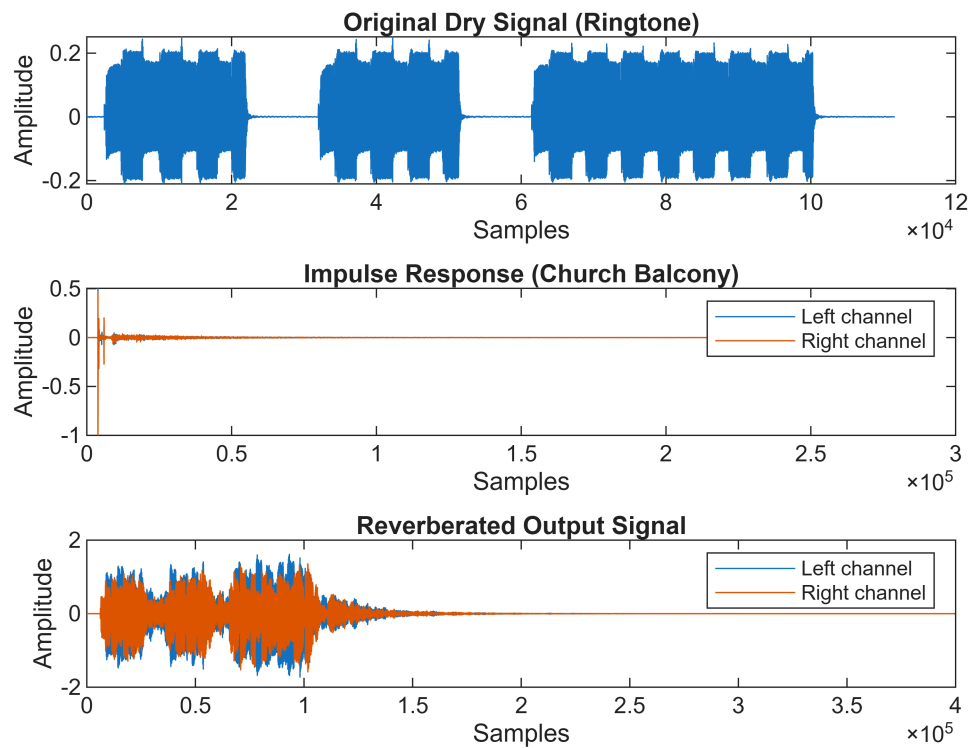
%% Plot the signals
figure;

subplot(3,1,1);
plot(x);
title('Original Dry Signal (Ringtone)');
xlabel('Samples'); ylabel('Amplitude');

subplot(3,1,2);
plot(h);
title('Impulse Response (Church Balcony)');
xlabel('Samples'); ylabel('Amplitude');
legend('Left channel', 'Right channel');

subplot(3,1,3);
plot(y);
title('Reverberated Output Signal');
xlabel('Samples'); ylabel('Amplitude');
legend('Left channel', 'Right channel');

```



Task 7

```
% MY OWN COSTOM FUNCTION:
function w = discrete_convo(u, v)
% without using MATLAB built-in conv()
Nu = length(u);
Nv = length(v);
% Output length
Nw = Nu + Nv - 1;
% Initialize output
w = zeros(1, Nw);
% Convolution sum
for n = 1:Nw
    for k = 1:Nu
        % Check overlap condition
        if (n - k + 1 > 0) && (n - k + 1 <= Nv)
            w(n) = w(n) + u(k) * v(n - k + 1);
        end
    end
end
end
end
%TEST of function:
clc
clear
close all
```

```

%% Given signals
x = [1 2 3 4 5]; %  $0 \leq n \leq 4$ 
h = [1 2 1]; %  $-1 \leq n \leq 1$ 
nx = 0:4;
nh = -1:1;
%% Manual convolution
y_manual = discrete_convo(x, h);
%% Built-in convolution (for comparison only)
y_builtin = conv(x, h);
%% Output index
ny = (nx(1) + nh(1)) : (nx(end) + nh(end));
%% Display results
disp("Manual convolution result:")

```

Manual convolution result:

```
disp(y_manual)
```

```

1      4      8     12     16     14      5

```

```
disp("Built-in conv result:")
```

Built-in conv result:

```
disp(y_builtin)
```

```

1      4      8     12     16     14      5

```

```

%% Plotting
figure
subplot(3,1,1)
stem(nx, x, 'filled')
title('Input x[n]')
xlabel('n')
subplot(3,1,2)
stem(nh, h, 'filled')
title('Impulse response h[n]')
xlabel('n')
subplot(3,1,3)
plot(ny, y_manual, 'o-', 'LineWidth', 2)
hold on
plot(ny, y_builtin, '*--', 'LineWidth', 2)
title('Manual vs Built-in Convolution')
xlabel('n')
legend('Manual', 'Built-in')
axis([-2 5 0 20])
grid on % set axis as requested

```

